

PROJECT PROGRESS FOR RESEARCH PROJECT

IIT 474-6

AI-Based Tomato Ripening Stage Detection and Sorting System with Shelf-Life Estimation for the Commonly Consumed *Padma* Variety in Sri Lanka Using Computer Vision and Conveyor Automation

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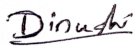
Declaration

We hereby declare that the research proposal titled

AI-Based Tomato Ripening Stage Detection and Sorting System with Shelf-Life Estimation for the Commonly Consumed *Padma* Variety in Sri Lanka Using Computer Vision and Conveyor Automation

is our original work and has been conducted solely for the purpose of fulfilling the partial fulfilment of the requirement of the course unit **IIT 474-6**. We affirm that this work is free from any form of plagiarism and has not been submitted, in part or in full, for any other course, program, or qualification. All sources used in this proposal have been properly cited and acknowledged.

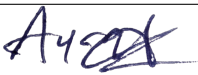
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List of Abbreviations

Artificial Intelligence	AI
Machine Learning	ML
Convolutional Neural Network	CNN
Artificial Neural Network	ANN
You Only Look Once version 8	YOLOv8
Mean Average Precision	mAP
Intersection over Union	IoU
Distribution Focal Loss	DFL
Frames Per Second	FPS
Graphics Processing Unit	GPU
Key Performance Indicator	KPI
Internet of Things	IoT
Pulse Width Modulation	PWM
Direct Current (motor)	DC
Light Emitting Diode	LED
Infrared (sensor)	IR
Universal Serial Bus	USB
High Definition (camera)	HD
Structured Query Language Lite	SQLite
Powerful Microcontroller Board	ESP32
Most widely cultivated and commonly consumed tomato variety (Sri Lanka)	<i>Padma</i>
Near Infrared	NIR

1 Introduction

1.1 Project Background

Tomato (*Solanum lycopersicum* L.) is one of the most economically important vegetable crops in Sri Lanka. The variety called *Padma* is widely grown in Sri Lanka because it tolerates tropical climate, gives good yield, and is accepted in both retail and wholesale channels. The stage of ripeness at harvest strongly affects flavour, firmness, shelf life, and selling price (Kumar et al. 2023). After harvest, tomatoes are typically graded by hand based on colour and visible surface condition. This manual process is subjective, slow, and depends on the experience and attention level of the worker. The FAO has reported that post-harvest losses in sorting and packaging can reach up to 20% in developing countries (FAO 2022).

With modern computer vision and deep learning, it is now possible to automate grading, defect detection, and conveyor-based sorting in real time (Li & Zhao 2022). In practical post-harvest settings, YOLO-style object detectors are particularly suitable because they can localize fruit instances in a full image and simultaneously assign class labels with high speed. A recent study by Moya et al. (2025) demonstrated a conveyor-integrated tomato grading prototype using YOLOv8, where tomatoes on a belt were detected and classified into three classes (green, red, and damaged) with real-time operation and very high accuracy. This work serves as a key baseline for the feasibility of YOLOv8-based automated sorting.

In this project, we extend the grading granularity to match real packing requirements by defining five classes: four ripeness stages (Stage 1 to Stage 4) and a defect class (e.g., bruises, cracks, rot). The dataset is annotated and managed using Roboflow, and the final YOLOv8 model is trained as a multi-class detector so that each detected tomato is directly labeled as Stage 1, Stage 2, Stage 3, Stage 4, or Defect. This single-model approach simplifies deployment because one network performs both localization and class prediction, which is beneficial for real-time conveyor operation.

Separately, shelf-life studies have shown that tomatoes harvested at different ripeness stages remain marketable for different durations. Controlled experiments link firmness degradation, colour change, and visible surface condition over time to remaining market life, and also model spoilage progression using physical or sensor-based measurements (Naim et al. 2024, Tarlak & Akbas 2023). However, much of this work focuses on controlled laboratory conditions and offline assessment, and does not integrate real-time visual grading with automated sorting

decisions or local packhouse conditions typical of developing-country post-harvest operations (Abekoon et al. 2024).

Therefore, this research targets two practical opportunities: (i) adapting real-time AI grading and sorting to a locally important Sri Lankan variety (*Padma*), and (ii) using visually detected grade (ripeness and defect) to support shelf-life estimation and operational decision-making. This project aims to address both opportunities by building an integrated conveyor-based system that performs YOLOv8-based five-class grading, defect rejection, shelf-life estimation, and KPI-driven decision support for the *Padma* variety under Sri Lankan post-harvest conditions. (Moya et al. 2025, Abekoon et al. 2024)

1.2 Problem Definition

This project addresses four core problems:

1. **Real-time grading for Padma tomatoes (5 classes):** How can a YOLOv8-based computer vision model accurately detect tomatoes and classify each instance into five categories (Stage 1–Stage 4 and Defect) in real time on a moving conveyor system?
2. **Defect detection and sorting automation:** How can defective tomatoes (e.g., bruised, cracked, or rotted) be identified as a separate class and physically rejected using a mechanical diverter while ripeness stages are routed to their respective collection bins?
3. **Shelf-life prediction:** How can the system estimate remaining shelf life based on the visually detected ripeness stage and observable external quality, so that packing and dispatch decisions can be prioritized? Prior work on shelf-life modelling typically employs laboratory-based measurements (temperature, humidity, firmness degradation, microbial growth) under controlled storage conditions, and does not integrate visual grade with real-time sorting or address local variety-specific spoilage patterns (Naim et al. 2024, Tarlak & Akbas 2023, Abekoon et al. 2024).
4. **Operational decision support:** How can we collect and visualize Key Performance Indicators (KPIs) such as throughput rate, five-class distribution, defect ratio, and predicted shelf-life statistics to assist operational and marketing decisions?

Existing work shows that CNN and YOLO-based approaches can grade tomatoes with high accuracy, and YOLO-based systems are well suited for real-time detection and sorting pipelines

(Rahman et al. 2021, Li & Zhao 2022, Moya et al. 2025). Huang et al. (2025) further improved YOLO-based tomato ripeness detection using multi-scale feature fusion and a small-object detection head, showing robust performance under challenging conditions such as overlap. However, most published systems:

- were trained on non-local tomato varieties,
- used only 2–3 ripeness categories (or did not include a clear defect-rejection class),
- did not integrate shelf-life estimation linked to visual grade in a live conveyor workflow.

There is very limited work targeting Sri Lankan post-harvest practice with local lighting, local handling conditions, and a locally important variety such as *Padma*. (Abekoon et al. 2024) **Therefore, this research is significant and worth undertaking** because it addresses the gap between existing AI-based tomato grading research and the operational needs of Sri Lankan post-harvest handling. By focusing on *Padma*, adopting a YOLOv8-based five-class grading model (Stage 1–Stage 4 plus Defect) trained using Roboflow-annotated data, and integrating shelf-life estimation with real-time conveyor sorting and KPI monitoring, this project aims to reduce post-harvest losses and enable data-driven packing and dispatch decisions. (Moya et al. 2025, Abekoon et al. 2024)

1.3 Project Aim and Objectives

Aim

To design and develop an AI-based real-time system to detect, classify, and sort **Padma** tomatoes based on ripeness stage and visible defects, and to estimate remaining shelf life.

Specific Objectives

- To collect and label a local dataset of more than 5,500 **Padma** tomato images covering four ripeness stages and one defect class under real farm and post-harvest lighting conditions.
- To develop and train a deep learning model that performs multi-class classification and defect detection.
- To design and construct a mechanical conveyor prototype mechanism that automatically sorts tomatoes according to detected ripeness stage and defects.
- To integrate and evaluate a shelf-life prediction and KPI monitoring system that estimates remaining freshness (in days) and displays class distribution, defect ratio, and throughput on a live dashboard.

2 Project Description

The proposed system is an AI-driven tomato grading and sorting prototype that integrates computer vision with conveyor-based mechanical actuation. The system:

- Detects tomatoes in real-time conveyor frames and assigns each detection to one of five classes (Stage 1–Stage 4 and Defect) using a YOLOv8 model.
- Routes tomatoes into four ripeness-based collection bins (Stage 1–Stage 4).
- Rejects tomatoes classified as *Defect* into a separate waste or rejection container.
- Estimates remaining shelf life based on the detected ripeness stage and post-harvest observation results.
- Logs Key Performance Indicators (KPIs) for operational monitoring and decision support.

A high-resolution camera (e.g., USB HD camera) is mounted above a motorised conveyor. Video frames are processed in real time using a YOLOv8-based detector trained on a Roboflow-annotated dataset. Each tomato is localized with a bounding box and classified directly into one of five classes (Stage 1–Stage 4 and Defect). This single-model approach supports practical deployment because detection and grading are performed within the same inference step, enabling low-latency decision-making for conveyor actuation (Moya et al. 2025).

2.1 Ripeness Stages

We will work with four ripeness stages plus a defect class, as summarized in Table 1.

Stage	Name	Visual Indicators
1	Green / Unripe	Uniform green colour, firm texture
2	Breaker	Green-yellow transition, less than 30% red area
3	Turning-Red	Orange to pink surface, 30–80% red coverage
4	Red / Fully Ripe	More than 80% red surface, softer texture
–	Defect	Cracks, bruising, and rot

Table 1: Proposed tomato grading classes for the *Padma* variety (four ripeness stages and a defect class).

Tomatoes are sorted into four ripeness-based collection bins (Stage 1–Stage 4), while defective tomatoes are separated for rejection. This enables stage-specific handling decisions, such as short-term marketing for later stages and longer storage or transport planning for earlier stages.



Figure 1: A batch of *Padma* tomatoes showing different ripeness stages, sizes, and surface defects used for dataset collection.

Figure 2: Visual representation of the four ripeness stages of *Padma* tomatoes (Stage 1–Stage 4) and the defect class used in this study.

2.2 Real-Time Sorting and Control Flow

The YOLOv8 model outputs the predicted class label (Stage 1–Stage 4 or Defect) together with a confidence score for each detected tomato. This information is transmitted to a microcontroller (Arduino UNO or ESP32) for actuation control. When a tomato is detected, the controller schedules the actuator trigger based on conveyor speed and the distance between the camera and the sorting gate. The actuator then diverts the tomato into one of four ripeness bins

(Stage 1–Stage 4), while tomatoes classified as Defect are diverted to a separate rejection path.

This approach is similar in spirit to the conveyor-integrated YOLOv8 grading system demonstrated by Moya et al. (2025), but it is extended and adapted in the following ways:

- Classification into **four ripeness stages** (Stage 1–Stage 4) rather than coarse categories.
- Inclusion of an explicit **Defect** class for quality rejection.
- Focus on the locally cultivated *Padma* variety under Sri Lankan post-harvest conditions.
- Integration of **shelf-life estimation** linked to the predicted ripeness stage.
- Sorting into **four separate bins** for ripeness stages, with a separate rejection bin for defective fruit.

The system also generates and logs Key Performance Indicators (KPIs), including throughput rate (tomatoes per minute), five-class distribution across batches, defect ratio, and average estimated shelf life per ripeness stage, to support operational and marketing decisions (Moya et al. 2025).



Figure 3: High-level processing and actuation flow of the proposed tomato grading and sorting system.

3 Literature Review

Multiple studies confirm that deep learning can classify tomato ripeness with high accuracy. (Li & Zhao 2022, Rahman et al. 2021) In real-time post-harvest environments, YOLO-style detectors are widely used because they can localize fruits in full-frame images and assign class labels at high speed, enabling integration with conveyor sorting systems. (Moya et al. 2025, Huang et al. 2025)

In this study, the core grading model is implemented as a **single YOLOv8 multi-class detector** trained to predict five classes (Stage 1–Stage 4 and Defect) directly from full-frame images. This design choice simplifies deployment by avoiding multi-model pipelines while still supporting fine-grained grading and defect rejection in real time. (Moya et al. 2025)

3.1 YOLO-Based Real-Time Systems

Moya et al. (2025) built a YOLOv8 conveyor system that achieved 99.6% classification accuracy and 97.1% size estimation accuracy using three classes (green, red, damaged). (Moya et al. 2025) This work demonstrates that conveyor-integrated real-time grading is technically feasible and commercially viable. (Moya et al. 2025) Patria et al. (2025) developed a vision-based mobile robot using Artificial Neural Networks (ANN) for tomato ripeness classification, showing that neural network approaches can be deployed beyond static conveyor setups and can scale to field robotics. (Patria et al. 2025)

3.2 CNN-Based Classification Approaches (Background)

CNN-based classifiers are widely used for tomato ripeness assessment, particularly when single-tomato crops are available and controlled imaging conditions are used. (Rahman et al. 2021, Allo et al. 2025) Allo et al. (2025) demonstrated the effectiveness of CNNs for multi-level tomato ripeness classification, highlighting their suitability for fine-grained maturity assessment beyond binary grading. (Allo et al. 2025) Centino et al. (2025) combined CNNs with image processing knowledge to improve interpretability and accuracy while maintaining efficiency. (Centino et al. 2025) Rahman et al. (2021) reported over 95% accuracy for tomato grading using CNNs, supporting CNN classifiers as robust choices for multi-class ripeness classification. (Rahman et al. 2021) Nalupano et al. (2024) and Ifmalinda et al. (2023) reported 90–94% accuracy for 3-class maturity classification using lightweight CNNs in controlled en-

vironments, indicating strong performance but highlighting the importance of field validation. (Nalupano et al. 2024, Ifmalinda et al. 2023)

3.3 Hybrid Detection + Classification Pipelines (Related Work)

Recent research also explores **hybrid approaches** that combine an object detector with a separate classifier, especially when very fine-grained grading is required or when the classifier is trained on carefully cropped samples. (Moya et al. 2025, Li & Zhao 2022) In a typical hybrid strategy:

- **YOLOv8 for detection:** Robust localization of multiple tomatoes in full-frame images with real-time performance. (Moya et al. 2025)
- **CNN for classification:** Fine-grained ripeness classification on cropped tomato regions with high accuracy. (Rahman et al. 2021, Allo et al. 2025)

Hybrid pipelines can offer benefits such as task separation and flexibility in training (detector and classifier can be optimized independently). (Moya et al. 2025) However, hybrid designs also increase deployment complexity and inference latency because they require cropping, a second network, and additional model management. Therefore, this study adopts a **single-model YOLOv8 multi-class approach** for direct five-class grading and real-time actuation compatibility. (Moya et al. 2025)

3.4 Enhanced YOLO Architectures and Multi-Scale Feature Fusion

Huang et al. (2025) proposed AITP-YOLO by adding a small-object detection head and multi-scale feature fusion to improve detection robustness in cluttered greenhouse images. (Huang et al. 2025) The model also runs efficiently on edge hardware (Jetson), addressing practical constraints of on-site deployment without high-end GPUs. (Huang et al. 2025) Liu et al. (2024) proposed multimodal fusion for tomato maturity assessment by combining multiple sensor modalities to improve robustness beyond single-channel image analysis. (Liu et al. 2024)

3.5 Deep Learning vs. Traditional Machine Learning

Liu et al. (2019) used traditional machine learning (colour features + SVM) to distinguish mature vs. immature tomatoes, achieving approximately 0.9 F1-score, but performance decreased

under changing light conditions and partial occlusion. (Liu et al. 2019) Garcia et al. (2019) used pixel-based color image classification with machine learning for tomato ripening stage identification, reporting reasonable accuracy in controlled laboratory settings but limited robustness for real packhouse environments due to lighting variability. (Garcia et al. 2019) These comparisons support the preference for CNN- and YOLO-based approaches in variable real-world conditions. (Li & Zhao 2022, Abekoon et al. 2024)

3.6 Color-Based and Spectroscopic Methods for Quality Assessment

Although deep learning dominates recent systems, complementary techniques remain useful for preprocessing and quality assessment. (Santoso et al. 2024, Borba et al. 2021) Santoso et al. (2024) compared color space representations (RGB, HSV, CMYK) using correlation-coefficient analysis for tomato ripeness classification, showing that color feature extraction can be useful as a baseline or supportive feature set. (Santoso et al. 2024) Borba et al. (2021) explored portable near-infrared (NIR) spectroscopy as a non-destructive method for fresh tomato quality control in field conditions, offering a complementary modality to visible-light imaging for improved assessment reliability. (Borba et al. 2021)

3.7 Summary of Supporting Articles Related to Tomato Ripeness Classification

Across the literature, YOLO-style detectors are repeatedly favored for real-time grading due to strong speed–accuracy trade-offs and suitability for conveyor integration. (Moya et al. 2025, Huang et al. 2025) CNN-based classifiers remain useful in settings where cropped single-tomato images are available and computational resources allow a second-stage classifier. (Rahman et al. 2021, Allo et al. 2025) Overall, the literature supports two viable directions (single-model YOLO vs. hybrid pipelines), and this project selects a **YOLOv8 multi-class model** to prioritize real-time deployment simplicity while still enabling five-class grading and defect rejection. (Moya et al. 2025)

Study	Method	Key Result	Limitations	Relevance to This Project
Moya et al., 2025	YOLOv8 on conveyor	99.6% accuracy; 97.1% size estimation	Only 3 classes (green/red/damaged); variety and local-condition coverage not explicit; no shelf-life module	Baseline for real-time sorting feasibility; motivates extending to 5-class grading + shelf-life + local Padma dataset.
Rahman et al., 2021	CNN (static images)	> 95% accuracy	Offline/static images; not validated on moving conveyor or real-time actuation; requires cropped inputs	Background benchmark for ripeness classification; supports deep learning but our work targets full-frame real-time grading.
Nalupano et al., 2024; Ifmalinda et al., 2023	Lightweight CNN (controlled)	90–94% accuracy (3-class)	Limited ripeness granularity (3 classes); controlled setting; no sorting hardware integration	Useful baselines; highlights need for multi-class + real-time conveyor evaluation.
Huang et al., 2025	AITP-YOLO (enhanced)	Improved clutter robustness; edge-friendly	Focuses on greenhouse scenes (not post-harvest conveyor); may require tuning for packhouse lighting	Guides robustness choices (occlusion/clutter) and edge deployment considerations.
Liu et al., 2024	Multimodal fusion	Improved robustness via multimodal sensing	Higher hardware cost/complexity; integration effort increases; may not suit low-cost setups	Suggests future enhancement beyond RGB; current scope prioritizes single-camera YOLOv8 deployment.
Zhang et al., 2018	CNN + robot	Vision + actuation integration	Different crop/setting; may not address fine-grained stages or post-harvest constraints	Supports end-to-end automation concept; informs actuator + vision integration.

Table 2: Supporting literature related to tomato ripeness classification and deployment.

3.8 Comparative Analysis of Tomato Classification Techniques

Method selection depends on whether the goal is single-image classification or real-time detection + grading in conveyor/field operation. (Moya et al. 2025, Li & Zhao 2022)

YOLO-based detectors dominate production systems because they:

- Detect and classify multiple tomatoes per frame simultaneously. (Moya et al. 2025)
- Run at real-time speeds on standard hardware. (Moya et al. 2025)
- Support integration with mechanical actuators. (Moya et al. 2025, Zhang et al. 2018)
- Scale from prototypes to packhouse deployments. (Moya et al. 2025)

CNN classifiers (ResNet family, VGG, etc.) are effective for:

- Cropped-image classification with high accuracy in controlled settings. (Rahman et al. 2021, Allo et al. 2025)
- Fine-grained multi-stage ripeness assessment when crops are reliable. (Allo et al. 2025, Rahman et al. 2021)
- Transfer learning and fine-tuning on local datasets. (Li & Zhao 2022)

Technique	Speed	Complexity	Accuracy (Ctrl.)	Robustness	Best For
YOLOv8 (multi-class detection)	30+ FPS	Low	High (multi-class)	Good	Real-time grading + sorting in one model
CNN (classification only)	10–20 FPS	Medium	High (multi-class)	Very good	Fine-grained grading on cropped images
Hybrid (YOLO + CNN)	15–25 FPS	Medium–High	High (multi-class)	Very good	When separate detector+classifier is required
Lightweight CNN	5–10 FPS	Medium	90–94%	Moderate	Resource-limited setups
Color + ML	Real-time	Very low	80–90%	Poor	Baseline grading / preprocessing
Multimodal	Varies	High	97%+	Excellent	Premium quality control

Table 3: Comparison of major tomato grading approaches.

3.9 Dataset Annotation and Preparation with Roboflow

Roboflow is used as the annotation platform for preparing the YOLOv8 training dataset. (Moya et al. 2025) In this study, a **single multi-class detection dataset** is created: full-frame images are annotated with bounding boxes around each visible tomato, and each bounding box is labeled as Stage 1, Stage 2, Stage 3, Stage 4, or Defect. The dataset is then exported in YOLO format for training a YOLOv8 multi-class detector. (Moya et al. 2025)

Data augmentation includes rotation, horizontal flip, brightness/contrast adjustment, blur, and noise injection. (Moya et al. 2025) The dataset split is 70% training, 20% validation, and 10% testing. (Moya et al. 2025)

3.10 Shelf-Life Prediction and Post-Harvest Management

Shelf-life prediction research combines visual ripeness assessment with time-based observation and sometimes environmental sensors to forecast remaining marketable days. (Li et al. 2024) Geetha et al. (2025) proposed deep learning regression for shelf-life prediction (days remaining) based on visual maturity stage and visible condition. (Geetha et al. 2025) Waseem et al. (2025) demonstrated rapid maturity prediction using image recognition and highlighted links between visual features and post-harvest quality trajectories. (Waseem et al. 2025)

3.11 Image Processing and Quality Control Under Field/Market Conditions

Field-based evidence shows a clear gap between laboratory evaluation and deployment under real conditions. (Abekoon et al. 2024) Key challenges include lighting variation, background clutter, handling damage, occlusion/overlap, and seasonal variation in ripening appearance. (Abekoon et al. 2024)

3.12 Identified Research Gaps

Although prior work demonstrates that deep learning can achieve high accuracy in tomato grading, several practical gaps remain for real-world post-harvest deployment. Many studies still rely on limited ripeness granularity (typically 2–3 classes), which is insufficient for fine-grained operational decisions such as packing priority and dispatch planning. In addition, a large portion of the literature evaluates models in controlled or offline settings, while relatively few studies validate performance on moving conveyors under realistic lighting varia-

tions, occlusion/overlap, and handling-related defects. End-to-end integration with mechanical actuation is also not consistently addressed, and shelf-life prediction is rarely incorporated into real-time sorting pipelines. Furthermore, operational decision support—such as KPI logging and monitoring dashboards—is often absent in academic prototypes, despite its importance for packhouse monitoring and process optimization.

These gaps motivate the proposed system, which focuses specifically on the Padma tomato variety, a widely cultivated local variety in Sri Lanka. The system targets five-class grading (Stage 1–Stage 4 and Defect) using a single YOLOv8 model, conveyor-compatible actuation, and KPI-based monitoring, with shelf-life estimation directly linked to the predicted ripeness stage to support local post-harvest handling requirements.

3.13 Novelty of This Project

This project presents a system-level design that integrates **YOLOv8-based five-class tomato grading** (Stage 1–Stage 4 and Defect), mechanical actuation for automated sorting, shelf-life estimation linked to ripeness stage, and KPI-based operational monitoring. (Moya et al. 2025, Zhang et al. 2018, Geetha et al. 2025) Unlike prior studies (Moya et al. 2025, Zhang et al. 2018, Geetha et al. 2025), the proposed system is specifically developed and evaluated using the *Padma* tomato variety, a widely cultivated local variety in Sri Lanka, addressing variety-specific post-harvest characteristics that are not explicitly considered in existing systems. (Abekoon et al. 2024)

3.14 System Resources and Implementation Approach

Hardware and software selections for the proposed system are aligned with successful real-time grading and automation studies. (Moya et al. 2025, Patria et al. 2025, Huang et al. 2025)

Category	Items / Tools	Purpose	Literature
Hardware	Conveyor belt, servo/pneumatic arm, LED lighting, camera	Real-time capture and physical sorting	Moya et al. (2025); Patria et al. (2025)
Software	Python, PyTorch, YOLOv8 (Ultralytics), Flask, SQLite, Roboflow	YOLOv8 training, real-time inference, dashboard, KPI logging	Huang et al. (2025); Moya et al. (2025)
Dataset	Locally captured <i>Padma</i> images with bounding-box labels (5 classes)	YOLOv8 full-frame multi-class detection dataset	Moya et al. (2025)
Other Tools	3D-printed arm, wooden frame, Arduino IDE, multimeter	Prototype construction and testing	Zhang et al. (2018)

Table 4: Summary of required resources for the proposed YOLOv8-based sorting system.

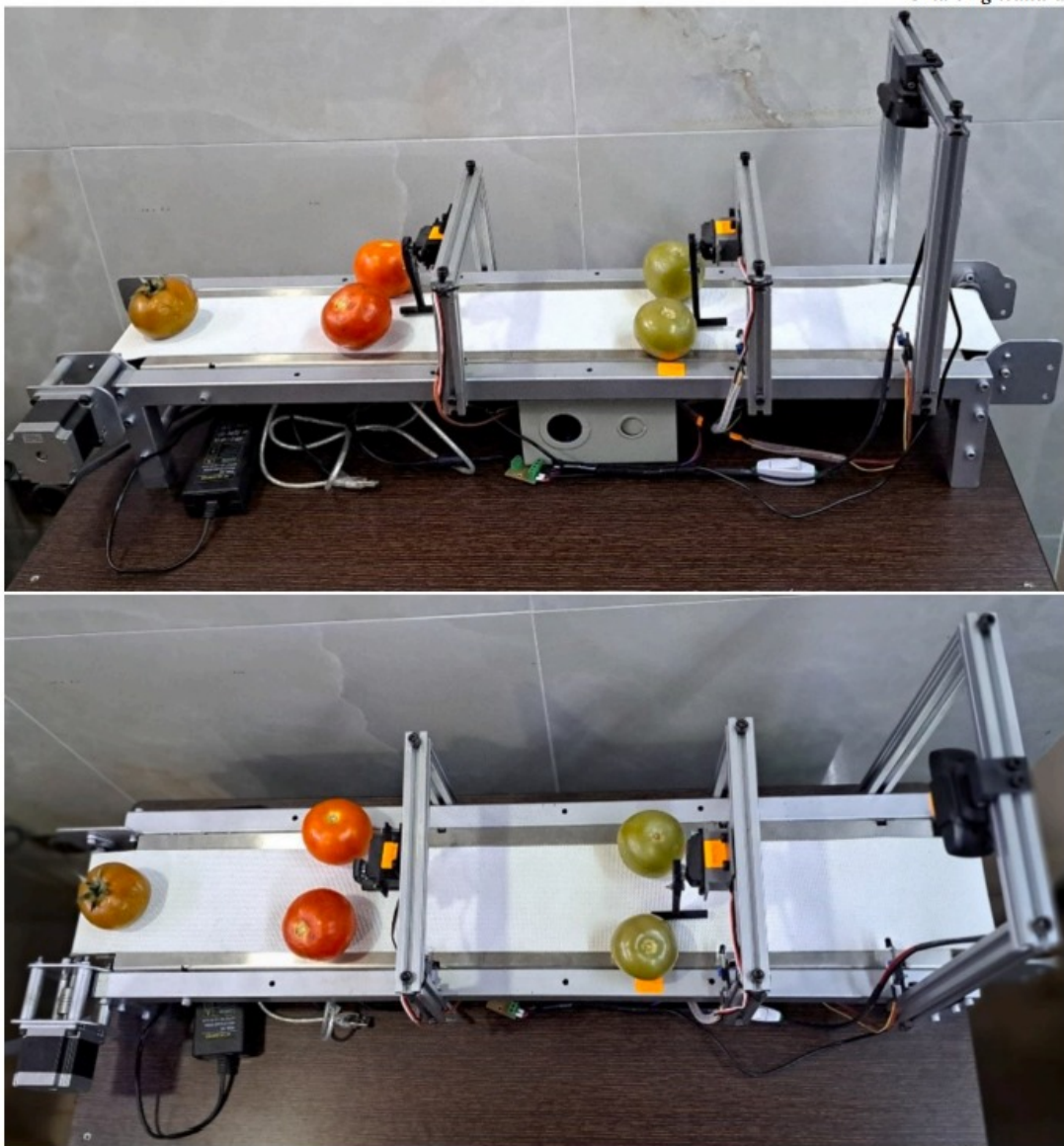


Figure 4: Existing YOLOv8-based conveyor prototype from prior research, which inspires the proposed design from (2025) Moya et al. (2025).

4 Methodology

The methodology follows a design-and-implementation pipeline with six main stages: data acquisition, dataset preparation, model development, hardware integration, software design, and testing/evaluation. The approach is experimental and iterative: we measure performance, refine the design, and retest until the target accuracy and real-time stability are achieved. (Moya et al. 2025, Abekoon et al. 2024)

4.1 Overall Approach

The proposed system performs an end-to-end closed-loop process: (i) capture tomato images/video frames on a moving conveyor, (ii) detect tomato instances and classify each instance directly into one of five classes (Stage 1–Stage 4 and Defect) using a YOLOv8 multi-class detector, (iii) estimate remaining shelf life based on the predicted ripeness stage and post-harvest observations, (iv) trigger an actuator to physically sort tomatoes into four ripeness bins (Stage 1–Stage 4) while rejecting Defect tomatoes into a separate container, and (v) log Key Performance Indicators (KPIs) for monitoring and decision-making. This integrates computer vision, deep learning, and mechatronics into one deployable pipeline. (Moya et al. 2025)

This pipeline is similar in spirit to the conveyor-integrated YOLOv8 system of Moya et al. (2025), but extended for the Sri Lankan *Padma* variety with four ripeness stages plus a defect class, and a shelf-life estimation module. The system is designed to operate in realistic post-harvest conditions as highlighted by Abekoon et al. (2024), while using a controlled imaging setup during data collection to reduce lighting variability and improve label consistency. (Abekoon et al. 2024)

Quantitative evaluation will include YOLOv8 detection/classification performance (e.g., mAP, precision, recall), actuator response time, sorting success rate, and operational throughput (tomatoes per minute). Robustness will be assessed under different belt speeds, tomato orientations, and moderate lighting variations. (Moya et al. 2025)

4.2 Data Acquisition

Collection Site and Sampling

Images will be collected at the **CLD Farm Fresh collecting center, Bandarawela**, where *Padma* tomatoes arrive from multiple cultivation areas in the Badulla region. This helps to cap-

ture natural variation in fruit size, surface texture, maturity progression, and handling-related defects while keeping the capture setup consistent. (Abekoon et al. 2024)

Controlled Capture Setup (White Box + LED)

To reduce background clutter and color distortion, image capture will be performed inside a **white capture box** with stable **LED lighting**. This controlled setup improves label consistency for ripeness stages and reduces the effect of unpredictable ambient lighting commonly present in collecting centers and packhouses. (Abekoon et al. 2024)

Dataset Size Target

We plan to capture **more than 9,600 images** of *Padma* tomatoes across four ripeness stages and a defect class. The dataset will include variation in tomato orientation, distance to camera, and mild occlusion to improve real-world generalization. (Moya et al. 2025)

4.3 Dataset Preparation

Class Definitions

Each tomato instance will be assigned one of the following labels:

- Stage 1 (Green / Unripe).
- Stage 2 (Breaker).
- Stage 3 (Turning-Red).
- Stage 4 (Red / Fully Ripe).
- Defect (cracks, bruises, rotting, or visible fungal infection).

Annotation Strategy (Single YOLOv8 Dataset via Roboflow)

All images will be annotated using Roboflow by drawing **bounding boxes** around each visible tomato and assigning one of the five class labels (Stage 1–Stage 4, Defect). The annotated dataset will be exported in YOLO format and used to train a YOLOv8 multi-class detection model. This single-dataset strategy is aligned with the objective of real-time deployment, because one model performs both localization and grading in a single inference step. (Moya et al. 2025)

Augmentation

To reduce overfitting and improve generalization, we will apply geometric data augmentation during training using random rotations and flips. Specifically, images will be rotated within a small angle range and horizontally flipped (and, if appropriate for the conveyor/camera setup, vertically flipped) to expose the model to realistic orientation variations.

Dataset Split

Data will be split into:

- Train: 70%.
- Validation: 10%.
- Test: 30%.

This split is chosen to provide a sufficiently large training set while keeping a relatively large held-out test set for unbiased performance evaluation, and class balancing will be enforced to avoid bias toward Stage 4 (fully ripe) and to ensure sufficient samples for early-stage classes and defects.

Unlike (Moya et al. 2025), who used a 70/20/10 train/validation/test split, this study adopts a 70/10/30 split to increase the size of the test set.

4.4 Model Development

Architecture Selection

A single-stage architecture will be deployed:

- **YOLOv8** for real-time tomato detection and five-class grading (Stage 1–Stage 4 and Defect) directly from full-frame images.

YOLOv8 is selected because it provides strong real-time detection performance and is suitable for conveyor environments, as demonstrated in conveyor-integrated tomato sorting studies. (Moya et al. 2025)

Training Process

Training will be conducted in Python using deep learning frameworks (e.g., PyTorch/Ultralytics YOLOv8). Planned steps include:

- **Optimiser:** default YOLOv8 optimiser settings (or Adam/SGD depending on results) with tuned learning rate.
- **Loss:** YOLOv8 detection losses (bounding-box regression, classification loss, and objectness loss) as provided by the framework.
- **Early stopping/model selection:** select the best checkpoint based on validation performance to reduce overfitting.

Model quality will be reported using precision, recall, and mean Average Precision (mAP) for the five classes, together with a confusion matrix for class-level error analysis. (Moya et al. 2025)

Ripeness and Defect Detection Module

The model will classify detected tomatoes into four ripeness stages and a defect class, enabling sorting decisions based on maturity and market acceptability. This is important for real post-harvest handling where defects must be rejected to maintain quality. (Abekoon et al. 2024)

4.5 Shelf-Life Prediction

A post-harvest observation experiment will be conducted to connect ripeness stage to remaining shelf life. Tomatoes with known ripeness stage (Stage 1–Stage 4) will be monitored under ambient conditions for several days, and the time at which each tomato becomes unsellable will be recorded (e.g., excessive softening, fungal/rotting development, severe blemishes). This study design is consistent with the need to evaluate quality under realistic market-like conditions rather than only controlled laboratory settings. (Abekoon et al. 2024)

Using these observations, shelf life will be estimated as *remaining shelf life in days* for each ripeness stage (and optionally used to define operational ranges). The predicted shelf-life value will be displayed together with the class label and logged for batch-level decision support. (Abekoon et al. 2024)

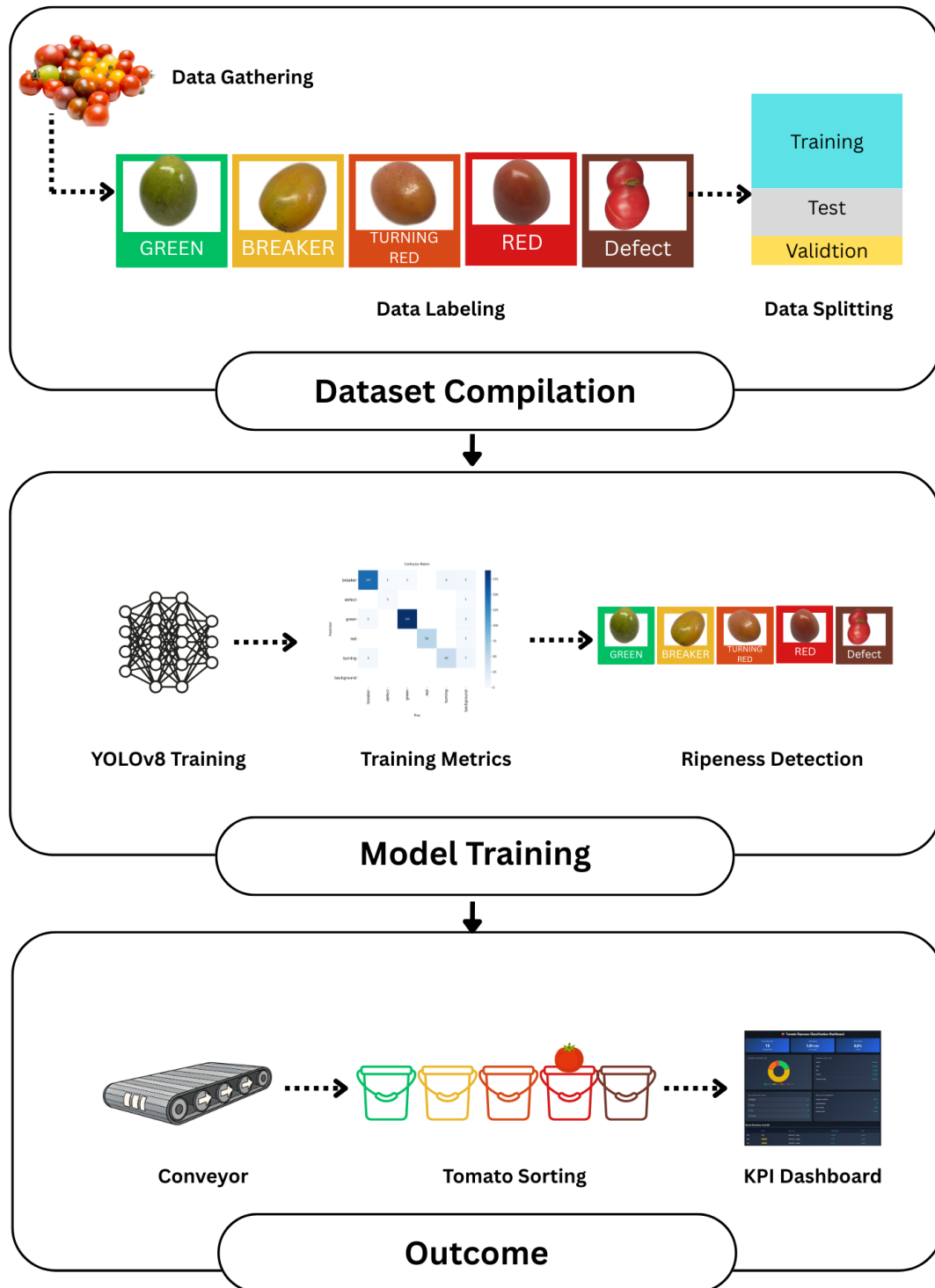


Figure 5: Overall methodology workflow of the proposed system: dataset compilation (data gathering, ripeness-stage labeling, and train/validation/test splitting), YOLOv8 model training and evaluation (training metrics), and deployment outcomes including conveyor-based sorting and KPI dashboard monitoring.

4.6 Hardware Integration

Hardware components:

- **Computing unit:** Laptop with NVIDIA graphics.
- **Conveyor system:** DC motor belt with PWM speed control.
- **Actuation:** Servo motor or pneumatic arm controlled via Arduino/ESP32.
- **Lighting:** LED strip with diffuser (plus the white-box capture system for dataset capture).
- **Sensing:** Proximity / IR sensor to detect tomato arrival under the camera and synchronize capture and actuation timing.

The real-time control flow is:

1. Camera captures a frame.
2. YOLOv8 detects tomatoes and outputs bounding boxes, predicted class (Stage 1–Stage 4 or Defect), and confidence.
3. Based on the predicted class, a control signal is sent to Arduino/ESP32 with timing synchronized to conveyor speed.
4. Arduino/ESP32 triggers the actuator to divert the tomato into the correct collection path: Stage 1–Stage 4 into four ripeness bins, and Defect into a rejection bin.
5. The system estimates shelf life (in days) based on the predicted ripeness stage and logs each event with KPIs.

4.7 Software Design

Processing Pipeline

- Frame acquisition (camera stream).
- Pre-processing (resize, normalize as required by the model).
- YOLOv8 inference (tomato detection + five-class prediction).

- Sorting decision logic and actuator timing control.
- Shelf-life estimation (days, based on stage mapping refined by observations).
- KPI logging to a local SQLite database and visualization via a Streamlit dashboard
- Live dashboard to show counts, five-class distribution, defect ratio, throughput, and shelf-life statistics.

KPI Analytics

Example KPIs:

- Defect Ratio = Defective tomatoes / total tomatoes.
- Sorting Throughput = Tomatoes sorted per minute.
- Class Distribution = Count(Stage 1), Count(Stage 2), Count(Stage 3), Count(Stage 4), Count(Defect).
- Estimated Remaining Shelf Life (average per batch, and per ripeness stage).

4.8 Testing and Evaluation

We will evaluate the system in four phases:

1. **Controlled environment tests:** White-box/LED lighting will be used to validate the vision model performance under consistent conditions and to fine-tune the image capture setup (camera mounting and frame rate).
2. **Stress tests:** Robustness will be tested by varying tomato orientation, partial occlusion, capture distance, and moderate lighting variation (simulated without conveyor operation).
3. **Field validation (future phase):** After prototype integration, pilot trials will be conducted in a realistic post-harvest/collecting-center environment using *Padma* tomatoes sourced from the collecting center supply chain.
4. **Benchmarking (future phase):** After end-to-end system completion, AI-based grading/sorting will be compared with manual grading in terms of time, consistency, and error rate.

4.9 Ethical and Environmental Considerations

This project exclusively uses images of tomatoes for analysis. No personal data, human subjects, or identifiable human information will be collected at any stage of the study. Access to the image dataset will be restricted to the research group members and academic supervisors only. Image acquisition is non-destructive, and tomatoes will not be intentionally damaged solely for the purpose of data collection (Abekoon et al. 2024).

Food waste will be minimized by directing usable tomatoes for normal consumption or processing, while spoiled samples will be disposed of responsibly after shelf-life observation. Electrical and mechanical components such as motors, actuators, and lighting systems will be operated within safe limits to prevent overheating and unsafe laboratory conditions. Where possible, low-power and energy-efficient hardware components will be used to reduce environmental impact (Abekoon et al. 2024).

The project supports sustainable agriculture by:

- reducing post-harvest loss through more consistent grading,
- enabling earlier shelf-life estimation to support marketing and packing decisions,
- improving sorting speed while reducing manual fatigue.

5 Results and Discussion

This section presents the outcomes achieved up to the progress evaluation stage of the project. At this stage, a **YOLOv8 multi-class detection model** has been trained and evaluated under a controlled image acquisition setup to detect and grade *Padma* tomatoes into five classes (green, breaker, turning, red, and defect). The current results therefore focus on model training behaviour, detection performance metrics (precision, recall, mAP), class-wise error patterns, and visual validation examples, while full conveyor–actuator integration and field trials are planned as the next development stage. (Moya et al. 2025, Abekoon et al. 2024)

5.1 Evaluation Dataset Summary

To reduce bias in evaluation, the dataset was prepared with near-balanced instance counts across the five classes. This helps to avoid inflated performance due to class imbalance and supports reliable class-wise error analysis.

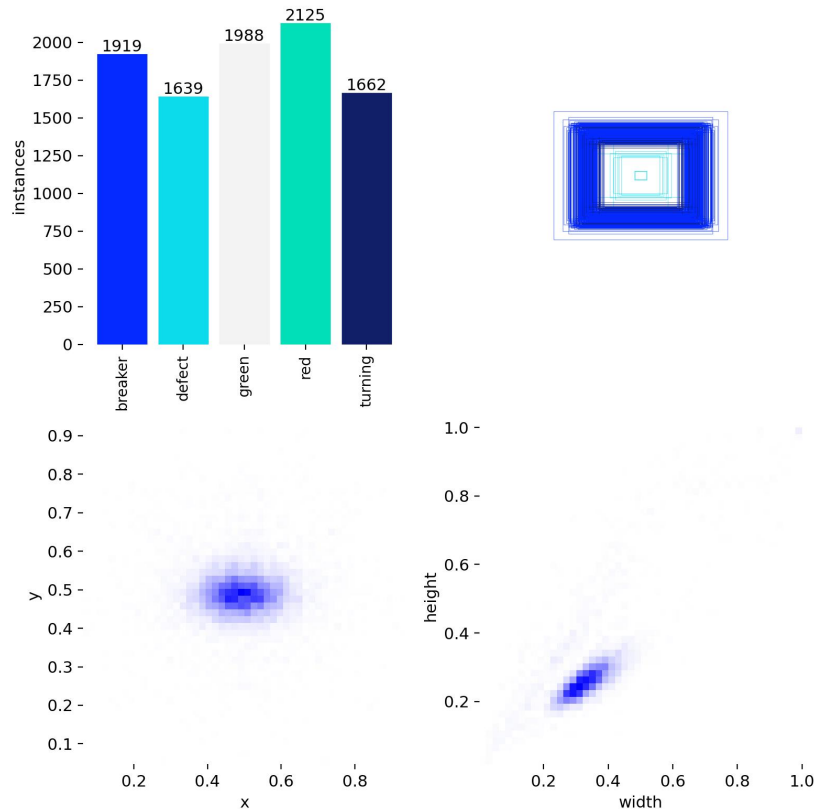


Figure 6: Dataset instance distribution and label statistics for five-class YOLOv8 training (green, breaker, turning, red, defect).

Figure 6 shows that the dataset contains comparable numbers of labeled instances per class,

enabling a meaningful comparison of errors between visually similar maturity stages such as *breaker* and *turning*, and between *defect* and non-defect samples.

5.2 Training Progress of the YOLOv8 Detector

Model training behaviour was monitored using training/validation losses (box loss, classification loss, and DFL loss) and detection metrics (precision, recall, mAP@0.5, and mAP@0.5–0.95). These curves are important because they indicate convergence stability, generalization behaviour, and potential overfitting.

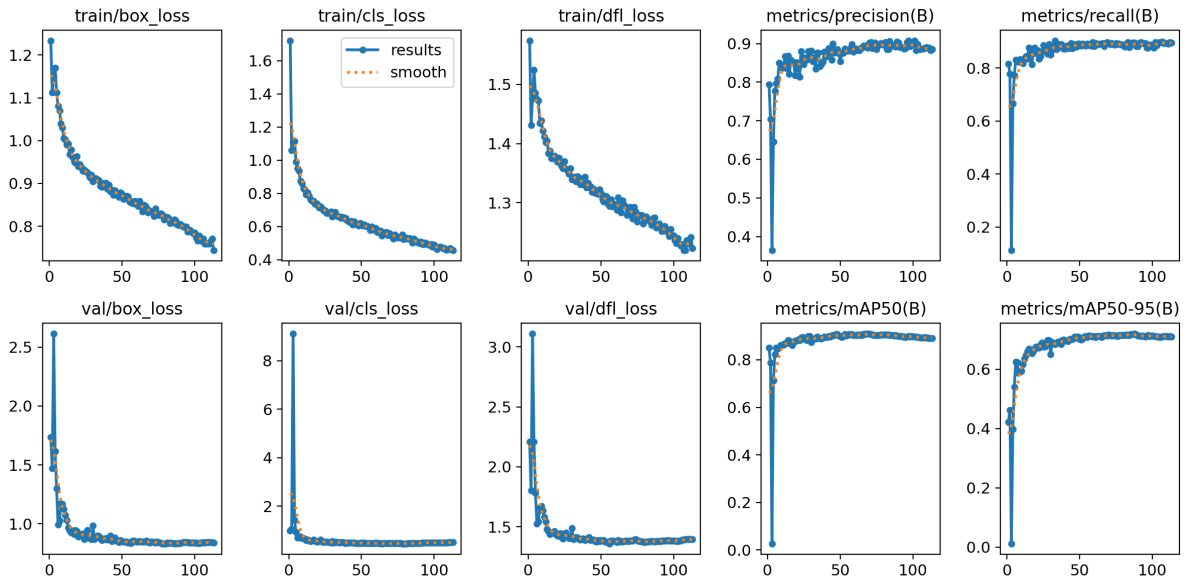


Figure 7: YOLOv8 training progress: training/validation losses and detection metrics across epochs.

As shown in Figure 7, both precision and recall rise rapidly during early epochs and later stabilize, while the loss curves decrease and remain stable, indicating effective learning under the controlled capture setting. In addition, mAP@0.5 reaches a high plateau, while mAP@0.5–0.95 remains lower as expected due to stricter IoU thresholds that penalize bounding-box localization quality more strongly.

5.3 Model Performance on Test/Validation Data

The final model performance was analysed using PR curves, confidence curves, and confusion matrices. These evaluations are useful because they show class-wise differences, decision-threshold trade-offs, and the exact confusion patterns between maturity stages.

Precision–Recall characteristics

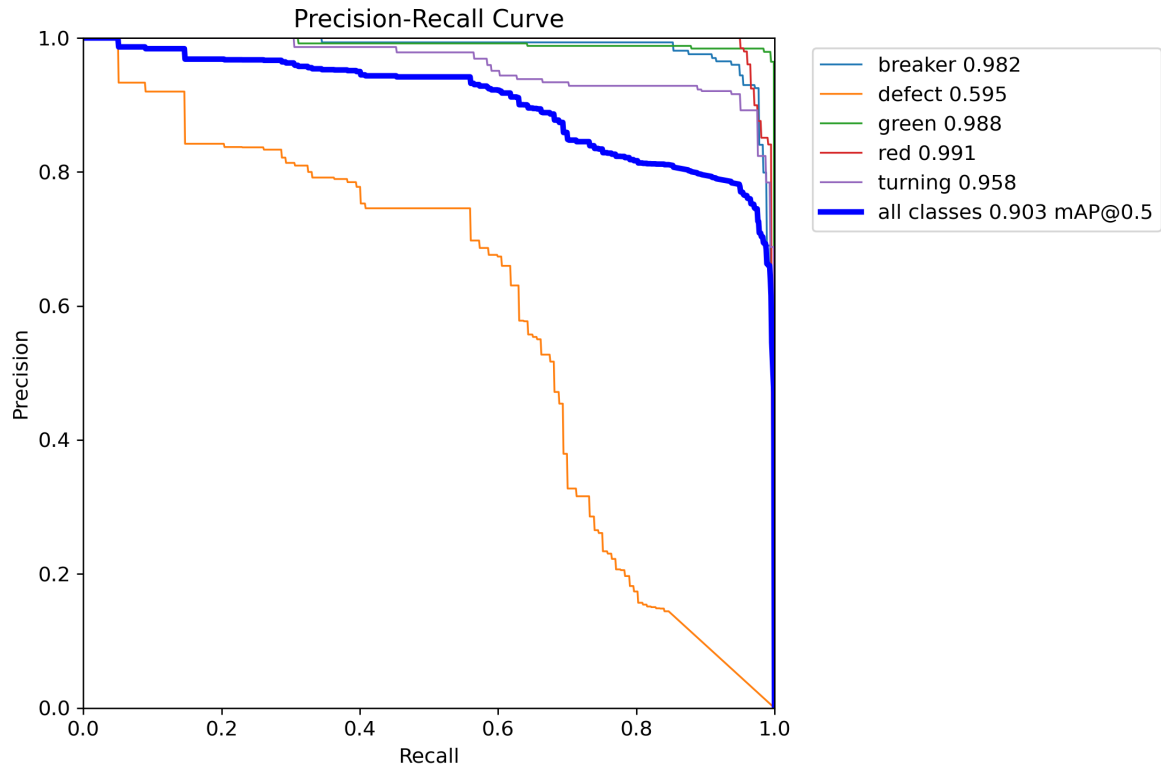
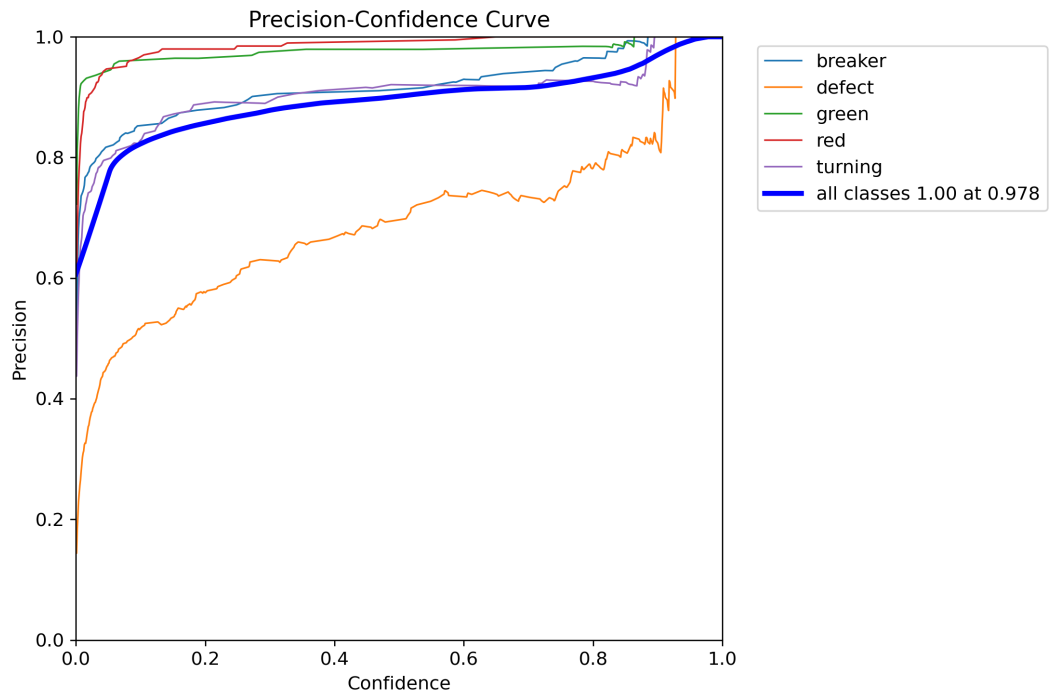


Figure 8: Precision–Recall curve for the five-class YOLOv8 detector (per-class and overall).

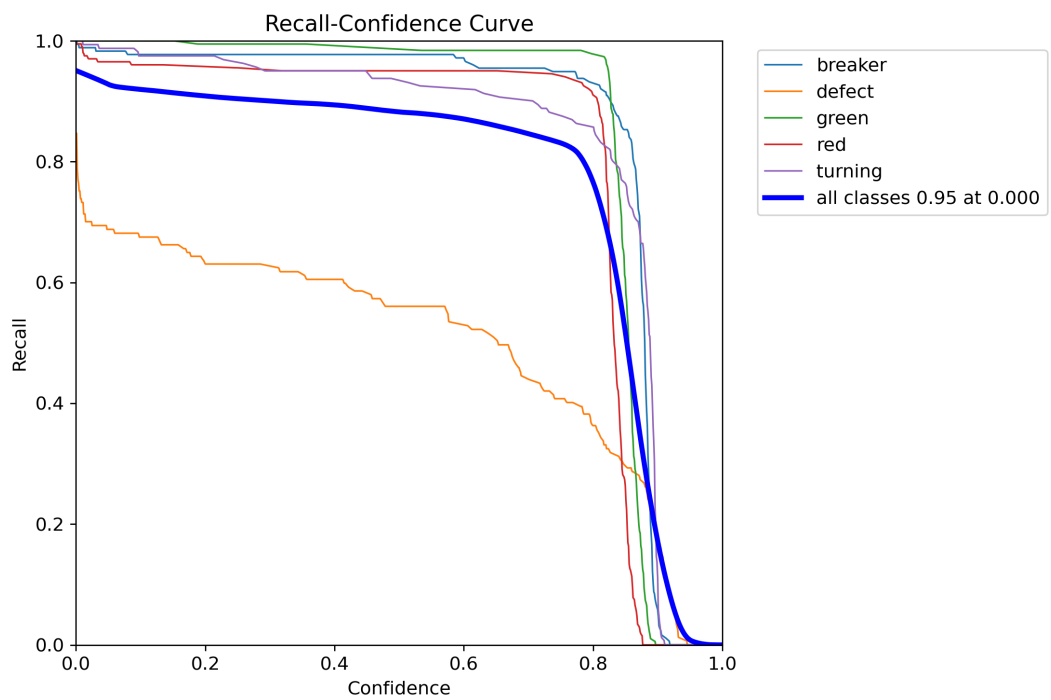
Figure 8 shows strong performance for visually distinctive classes (especially *green* and *red*), while the *defect* class yields noticeably weaker performance compared to the ripeness classes. This indicates that defect detection is the most challenging component and is likely to be the main source of errors in real deployment.

Confidence-threshold trade-off

Figures 9 and 10 illustrate the expected trade-off: as the confidence threshold increases, precision typically increases while recall decreases. This is particularly important for the *defect* class, where a higher threshold may reduce false positives but can also increase the risk of missing defective tomatoes, which is a critical operational concern.



(a) Precision vs. confidence.



(b) Recall vs. confidence.

Figure 9: Precision and Recall confidence-threshold trade-off curves for the five-class YOLOv8 detector.

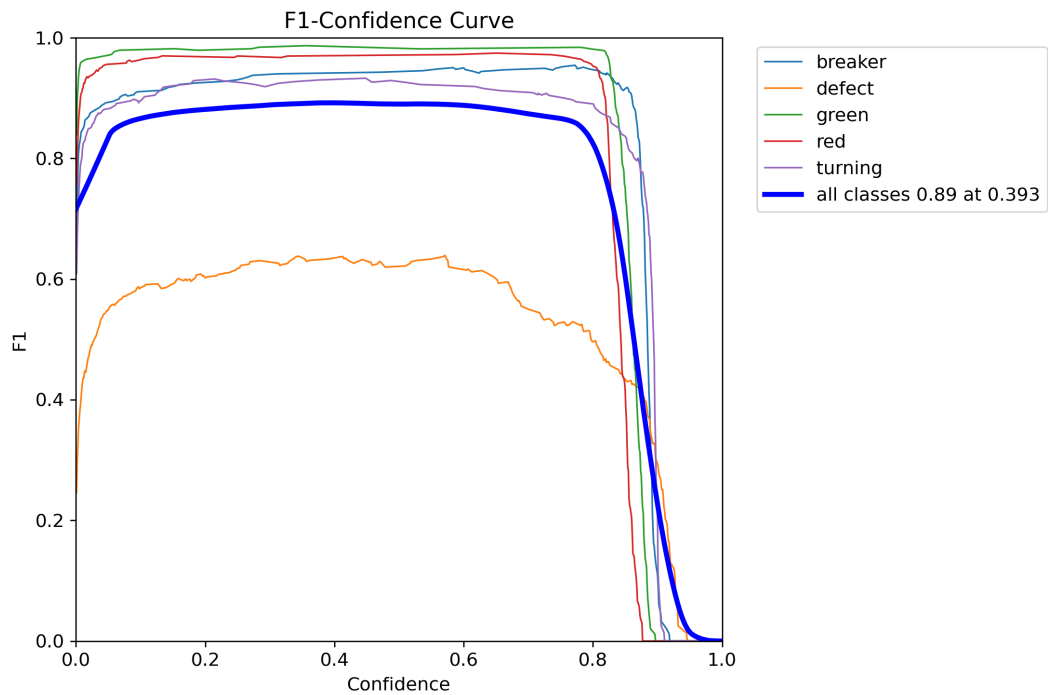


Figure 10: F1-score vs. confidence threshold for the five-class YOLOv8 detector (per-class and overall).

Confusion matrix analysis

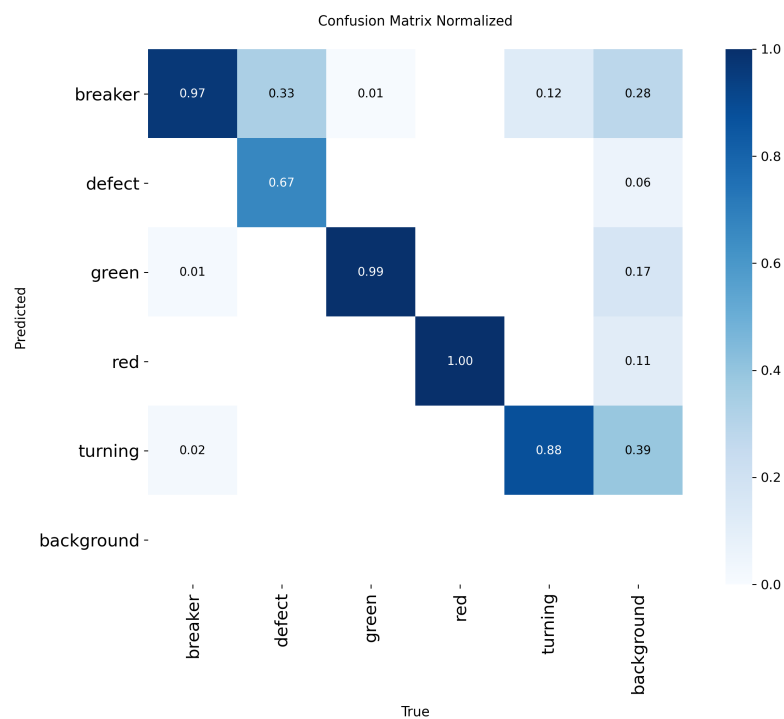


Figure 11: Normalized confusion matrix for five-class YOLOv8 predictions (including background).

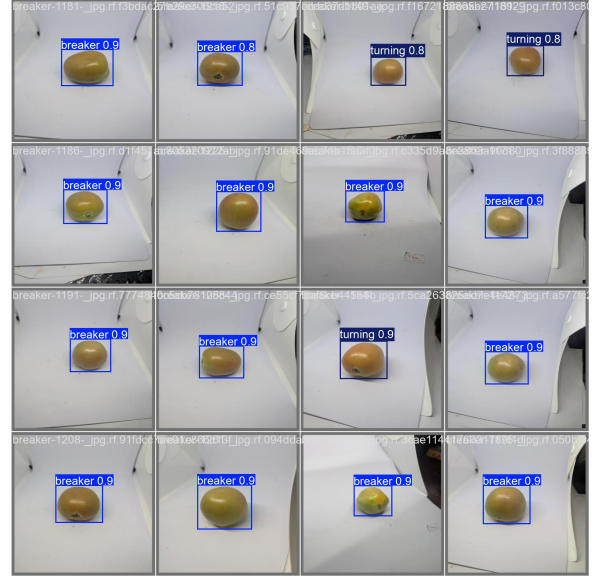
Figure 11 indicates that most predictions lie on the diagonal for ripeness classes, confirming strong class separability for *green* and *red*. Misclassifications occur mainly between adjacent maturity stages (e.g., *breaker* vs. *turning*), which is expected because these stages represent gradual color transitions rather than sharply separable visual categories. The defect class shows lower correctness relative to other classes, suggesting that defect appearance diversity (bruise, crack, rot, fungal spots) and small affected regions make defect learning more difficult.

5.4 Qualitative Validation (Detection Examples)

To complement quantitative metrics, qualitative evaluation was conducted using predicted bounding boxes on validation images. These visualizations help verify correct localization, identify missed detections, and reveal typical confusion patterns across the ripeness and defect classes.



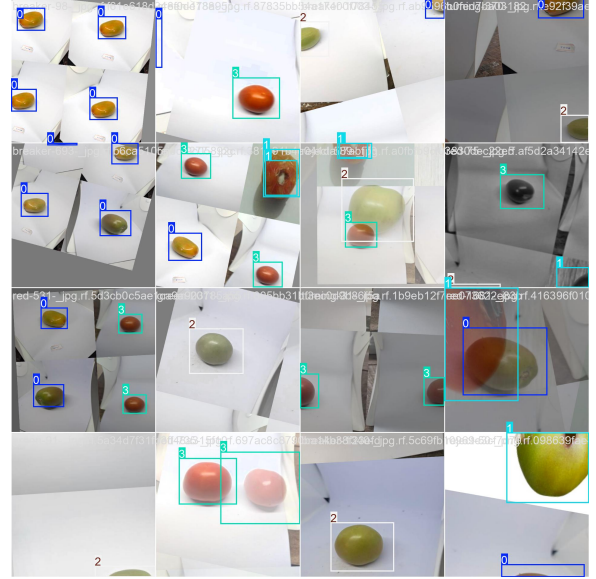
(a) Validation predictions (batch 0).



(b) Validation predictions (batch 1).



(c) Validation predictions (batch 2).



(d) Additional example prediction (train/val mosaic).

Figure 12: Qualitative YOLOv8 detection examples showing predicted bounding boxes and class labels across multiple batches.

Figure 12 shows that the model generally detects tomatoes correctly under the controlled white-box environment. However, occasional class ambiguity is observed for visually similar intermediate stages (breaker/turning) and for challenging defect appearances, which aligns with the confusion-matrix patterns.

5.5 Shelf-Life Estimation

This experiment presents the final outcome of shelf-life estimation of *Padma* tomatoes based on colour differentiation across four ripening stages: Green, Breaker, Turning Red, and Red. The study was conducted under room-temperature conditions over a 31-day time-series period.

Tomatoes were observed every morning, and changes in surface colour were used to identify the ripening stage. Images captured during the experiment illustrate the gradual transition of tomatoes through the four stages (Figure 14).

Based on the observations, the duration of each ripening stage was identified as 2 days (Green), 5 days (Breaker), 12 days (Turning Red), and 12 days (Red). Using these durations, the estimated remaining shelf life for each stage was calculated and is summarised in Figure 13.

The results show that tomatoes at earlier ripening stages have a longer remaining shelf life, while fully ripe tomatoes have a shorter remaining shelf life, supporting effective grading and post-harvest handling decisions.

Tomato Stage	Duration of Stage (Days)	Shelf Life (Days)
 Green	2	31
 Breaker	5	29
 Turning Red	12	24
 Red	12	12
Total	31	—

Figure 13: Estimated remaining shelf life of *Padma* tomatoes across four ripening stages based on observed stage durations under room-temperature conditions (31-day observation period).

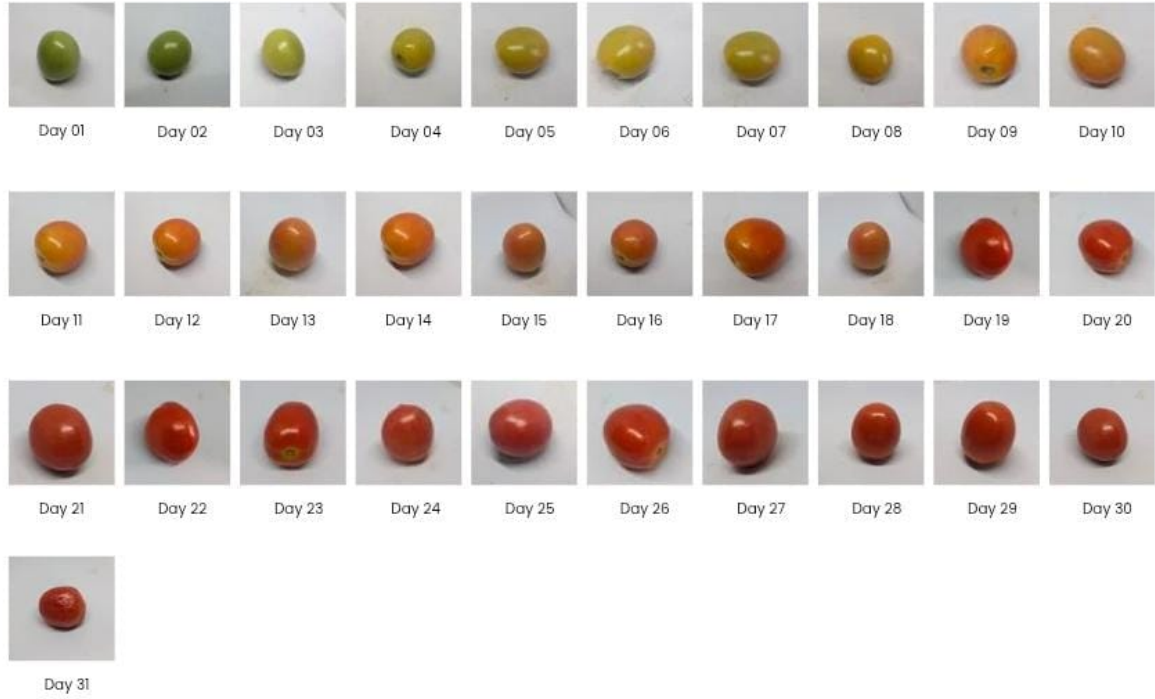


Figure 14: Visual colour differentiation of *Padma* tomato ripening stages observed over a 31-day period under room-temperature conditions.

5.6 Discussion: Key Observations, Limitations, and Next Steps

The current results confirm that YOLOv8-based five-class detection and grading is feasible under controlled imaging conditions, achieving strong overall detection metrics and stable convergence behaviour during training. The most frequent grading errors occur between *breaker* and *turning*, which reflects the continuous nature of ripening and suggests that label consistency and borderline-stage sampling are critical for improved reliability. In addition, defect detection remains the most challenging class and requires further dataset expansion and refinement before deployment in a rejection-critical conveyor system. (Abekoon et al. 2024)

At this progress stage, three limitations remain:

- **Conveyor-actuator evaluation is pending:** Real-time performance metrics such as throughput (tomatoes/min), actuator response time, and sorting success rate will be measured after full conveyor integration. (Moya et al. 2025)
- **Defect generalization needs improvement:** Current evidence indicates weaker defect precision/recall compared to ripeness classes; more diverse defect samples and stricter annotation rules are required for robust rejection decisions.

- **Shelf-life mapping remains baseline:** Shelf-life values are currently stage-based estimates and will be validated and refined using the planned post-harvest observation study under realistic local conditions. (Abekoon et al. 2024)

The next development milestone is to integrate the trained YOLOv8 model with the conveyor system, synchronize inference output with sensor timing, and evaluate end-to-end sorting (four ripeness bins plus defect rejection) using real throughput and sorting success metrics, aligning with real-time feasibility demonstrated in prior conveyor-based tomato sorting research. (Moya et al. 2025)

6 Individual Contribution and Work Plan

Enrolment No.	Name	Key Responsibilities
UWU/IIT/21/017	K.R.D.H. Gunawardhana	• Participate in data collection and labeling of the local Padma tomato dataset. • Lead shelf-life prediction system development and KPI monitoring integration. • Support prototype development and testing. • Contribute to documentation.
UWU/IIT/21/019	P.L.H. Hirushan	• Participate in data collection and labeling of the local Padma tomato dataset. • Lead the design and construction of the mechanical conveyor prototype. • Implement the diverter mechanism for sorting. • Support hardware integration and system evaluation. • Contribute to documentation.
UWU/IIT/21/062	W.K. Gayan	• Participate in data collection and labeling of the local Padma tomato dataset. • Lead the development and training of the Deep Learning model for ripeness and defect detection. • Integrate the model with hardware for real-time inference. • Support system testing and performance optimization. • Contribute to documentation and model evaluation reporting.

7 Project Timeline

The expected timeline along which the research is planned to proceed.

Months 1–4 (Nov 2025 – Feb 2026): Literature Review and Background Study

- Review vision-based tomato grading, deep learning model-based conveyor systems, and sorting automation.
- Finalise the problem scope for the *Padma* tomato variety.
- Define dataset requirements and hardware constraints.

Months 1–6 (Nov 2025 – Apr 2026): Dataset Collection and Preparation

- Capture *Padma* tomato images under real-world conditions (multiple ripeness stages and defects).
- Label each tomato with ripeness stage (1–4) and defect / non-defect.
- Augment data and organise train / validation / test splits.

Months 4–6 (Feb 2026 – Apr 2026): Model Training and Validation

- Train the deep learning model for multi-class ripeness and defect detection.
- Tune accuracy, precision, recall, and F1-score.
- Test real-time inference on live video or conveyor footage.

Months 5–8 (Mar 2026 – Jun 2026): Conveyor Design and Prototype Assembly

- Build the conveyor-based sorting rig with camera mount, lighting frame, and sensors.
- Implement actuator control (servo/pneumatic) via Arduino.
- Prepare physical housing and safety enclosure for operation.

Months 6–10 (Apr 2026 – Aug 2026): System Integration and Testing

- Deploy the trained model to the computer.
- Connect inference output to actuator control for physical removal of tomatoes (Stage 4 / defect).

- Debug the full loop: camera → AI model → actuator → logging.

Months 7–10 (May 2026 – Aug 2026): KPI Evaluation and Performance Analysis

- Measure throughput (tomatoes per minute), defect ratio, and classification accuracy.
- Record shelf-life predictions and compare with observed shelf-life tracking.
- Benchmark AI sorting versus manual grading in terms of speed and consistency.

Months 8–10 (Jun 2026 – Aug 2026): Documentation and Final Submission

- Prepare the final report, figures, and presentation materials.
- Summarise KPI dashboard results and system performance.
- Deliver the final demonstration of the working prototype.

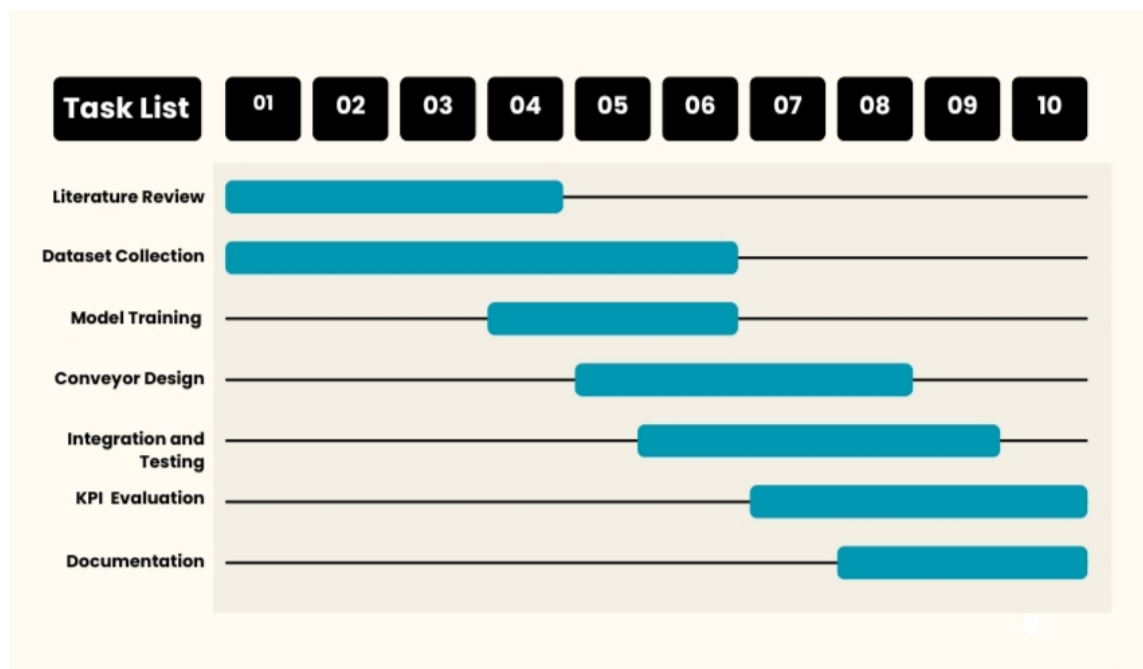


Figure 15: Gantt-style timeline of project phases (Nov 2025 – Aug 2026).

Project Milestones

Table 5: Project Milestones and Assigned Responsibilities

No.	Milestone / Task	Timeframe	Responsible Member(s)
M1	Literature review and background study. Define project scope, objectives, and methodology.	Nov 2025 – Feb 2026	UWU/IIT/21/017 UWU/IIT/21/019 (Lead) UWU/IIT/21/062
M2	Data collection: capture and label <i>Padma</i> tomato images for four ripeness stages and defects.	Nov 2025 – Apr 2026	UWU/IIT/21/017 UWU/IIT/21/019 UWU/IIT/21/062 (Lead)
M3	Dataset annotation, verification, and augmentation for model training.	Jan 2026 – Apr 2026	UWU/IIT/21/017 (Lead) UWU/IIT/21/019 UWU/IIT/21/062
M4	Deep learning model design, training, and optimization.	Feb 2026 – Apr 2026	UWU/IIT/21/019 (Lead) UWU/IIT/21/017 UWU/IIT/21/062
M5	Conveyor prototype design, actuator and sensor integration.	Mar 2026 – Jun 2026	UWU/IIT/21/062 (Lead) UWU/IIT/21/017 UWU/IIT/21/019
M6	Shelf-life experiment setup, data recording, and KPI analysis.	Apr 2026 – Aug 2026	UWU/IIT/21/017 UWU/IIT/21/019 (Lead) UWU/IIT/21/062
M7	System integration and performance evaluation (field testing, calibration).	May 2026 – Aug 2026	UWU/IIT/21/017 UWU/IIT/21/019 UWU/IIT/21/062 (Lead)
M8	Documentation, report preparation, and proposal refinement.	Jun 2026 – Aug 2026	UWU/IIT/21/017 (Lead) UWU/IIT/21/019 UWU/IIT/21/062

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